Assignment No: 1

HIDAYATULLAH

Electrical Ai

Semester 2nd

Section B

Registration BF25NWELE0685

```
#include <iostream>                    // Library for input/output
using namespace std;                   // Allows using cout, cin without std::

 // Built in Function to convert grade (A, B, etc.) into grade points
double getGradePoint(char grade) { // Function starts
   if (grade == 'A') return 4.0; // If grade is A, return 4.0
   else if (grade == 'B') return 3.0; // If grade is B, return 3.0
   else if (grade == 'C') return 2.0; // If grade is C, return 2.0
   else if (grade == 'D') return 1.0; // If grade is D, return 1.0
   else return 0.0; // For F or invalid grade, return 0
}

int main() {                                   // Main function starts

   // i. Declare variables
   double voltage, current, resistance;        // Declaring three double variables

   // ii. Display message
   cout << "Welcome to OOPs in C++!" << endl;  // Printing welcome message
```

```
// iii. Read age
int age;                                  // Declare integer variable for age
cout << "Enter your age: ";               // Ask user to enter age
cin >> age;                               // Take input from user

// iv. Temperature check
double temperature;                       // Declare temperature variable
cout << "Enter temperature: ";            // Ask for temperature
cin >> temperature;                       // Input temperature
if (temperature > 100)                    // Check if temperature is greater than 100
   cout << "Overheating detected!\n";     // Print warning

// v. Area of circle
double r, area;                           // Declare radius and area
cout << "Enter radius: ";                 // Ask for radius
cin >> r; // Input radius
area = 3.14159 * r * r;                   // Calculate area using formula πr²
cout << "Area of circle: " << area ;      // Display area

// vi. Print x, y, z
int x, y, z;                              // Declare three integers
cout << "Enter x, y, z: ";                // Ask user
cin >> x >> y >> z;                       // Input values
cout << x << "," << y << "," << z << endl; // Print values separated by comma

// vii. Sum of voltage and current
cout << "Enter voltage and current: ";    // Ask user
cin >> voltage >> current;                // Input voltage and current
double total = voltage + current;         // Calculate sum
cout << "Total: " << total << endl;       // Display result
```

```
// viii. Division check
double numerator, denominator;                    // Declare variables
cout << "Enter numerator and denominator: ";      // Ask user
cin >> numerator >> denominator;                  // Input values
if (denominator == 0)                             // Check if denominator is zero
   cout << "Error: Division by zero" << endl;     // Display error

// ix. Increment count
int count;                                        // Declare count
cout << "Enter count: ";                          // Ask user
cin >> count; // Input value
cout << "After increment: " << ++count << endl;   // Increment and print

// x. Read three floats
float a, b, c;                                    // Declare three float variables
cout << "Enter three numbers: ";                  // Ask user
cin >> a >> b >> c;                               // Input values

// xi. Series resistance
double r1, r2, r3;                                // Declare resistances
cout << "Enter three resistances (series): ";     // Ask user
cin >> r1 >> r2 >> r3; // Input values
cout << "Total Series Resistance: " << (r1 + r2 + r3) ;   // Formula R = r1 + r2 + r3

// xii. Parallel resistance
cout << "Enter three resistances (parallel): ";   // Ask user
cin >> r1 >> r2 >> r3; // Input values
cout << "Total Parallel Resistance: " << 1 / (1/r1 + 1/r2 + 1/r3) << endl;
                                                  // Formula for parallel
```

// xiii. Voltage division

```
double V, R1, R2;                                          // Declare variables
cout << "Enter V, R1, R2: ";                               // Ask user
cin >> V >> R1 >> R2;                                       // Input values
cout << "Voltage across R2: " << V * (R2 / (R1 + R2)) << endl;
                                                           // Voltage division formula
```

// xiv. Current division

```
double I;                                                  // Declare current
cout << "Enter total current I, R1, R2: ";                 // Ask user
cin >> I >> R1 >> R2; // Input values
cout << "Current through R2: " << I * (R1 / (R1 + R2)) << endl;
                                                           // Current division formula
```

// xv. GPA calculation using alphabet grades

```
char g1, g2, g3;                                        // Declare grade variables (A, B, etc.)
double ch1, ch2, ch3;                                   // Declare credit hours

cout << "Enter grades (A/B/C/D/F) and credit hours:\n"; // Ask user

cout << "Subject 1: ";                                  // Prompt for subject 1
cin >> g1 >> ch1;                                       // Input grade and credit hours

cout << "Subject 2: ";                                  // Prompt for subject 2
cin >> g2 >> ch2;                                       // Input grade and credit hours

cout << "Subject 3: ";                                  // Prompt for subject 3
cin >> g3 >> ch3;                                       // Input grade and credit hours

double gp1 = getGradePoint(g1);                         // Convert grade to points
double gp2 = getGradePoint(g2);                         // Convert grade to points
double gp3 = getGradePoint(g3);                         // Convert grade to points

double gpa = (gp1*ch1 + gp2*ch2 + gp3*ch3) / (ch1 + ch2 + ch3);
                                                        // GPA formula

cout << "Semester GPA: " << gpa << endl;                // Display GPA

return 0;                                               // End of program
}
```